

Introduction to Text-Analysis

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Research Interests:

text-analysis,
public economics

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Program:

- Why we need to learn text analysis
- What is text analysis: introduction to basic methods
- Text Normalization
- Text Analysis

Before to start: best friends

A common error

```
ValueError: math domain error
```



Sometimes we have to start from the past

"We are merely reminding ourselves that human decisions affecting the future, whether personal or political or economic, cannot depend on strict mathematical expectation, since the basis for making such calculations does not exist; and that it is our innate urge to activity which makes the wheels go round, our rational selves choosing between the alternatives as best we are able, calculating where we can, but often falling back for our motive on whim or sentiment or chance."

Three main point

- This quote was made in 1936.(JMK, The General Theory of Employment, Interest and Money)
- Today we have social media, blogs and millions of opinions.
- Today we have better computers.

Three main point II

Are we sure we can not use this information? Are we able to extract what people are currently thinking and better understand people's rationality?

- Text analysis (NLP or computational linguistics) is an umbrella term for a range of computational tools and statistical techniques that quantify text.
- Text is usually defined as unstructured data, as opposed to structured data (numbers). Are we sure?
- **Key text data:** news, articles, financial contracts, social media, supervisory and written reports.
 - → Web- Scraping!

What is text analysis?

Text-Analysis can be used to **understand** the content of a text *or/and* to **extract information** by a group of words.

What is text analysis? Content!

"And so we view this, and I do not think we are unbiased observers, we think the euro is irreversible. And it's not an empty word now, because I preceded saying exactly what actions have been made, are being made to make it irreversible. But there is another message I want to tell you. Within our mandate, the ECB is ready to do whatever it takes to preserve the euro. And believe me, it will be enough"

Computer-enabled

- To process and summarize far more text than a person is able to do.

Pros

- rule out subjectivity
- increase validity of the analysis

Cons

- Machines are (*still*) not able to understand emotions
- Communication relies also on who is talking.

What is text analysis? Extracting Information

EN:

This is a nice studio apartment with one bathroom for rent. It has a simple decor that makes it a functional and comfortable space. It receives natural light from a large set of windows. The living room has a single bed, a desk and a dining area with TV. The kitchen comes fully equipped with all of your cooking needs, plus a microwave, refrigerator, and stove. The bathroom is large. The building's common areas include a lounge, community kitchen, and a laundry room with washers and dryers. You'll also enjoy the courtyard.

You'll love living in Lichtenberg. Next to two major parks, Fennpfulpark and Stadtpark Lichtenberg, you'll enjoy spending your time outdoors. Plus, public transportation is easy with a tram and bus stop right around the block.

For groceries, there's a Lidl conveniently down the street!

What is text analysis? Extracting Information II

EN:

This is a nice studio apartment with one bathroom for rent. It has a simple decor that makes it a functional and comfortable space. It receives natural light from a large set of windows. The living room has a single bed, a desk and a dining area with TV. The kitchen comes fully equipped with all of your cooking needs, plus a microwave, refrigerator, and stove. The bathroom is large. The building's common areas include a lounge, community kitchen, and a laundry room with washers and dryers. You'll also enjoy the courtyard.

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For groceries, there's a Lidl conveniently down the street!

What is text analysis? Extracting Information III

	Our Fancy house in Lichtenberg
Park	1
Caffè	0
Cinema	0
Grocery	1
Farmacy	0
Public Transports	
Bus	1
Tram	0
Metro	1

Text-Analysis application

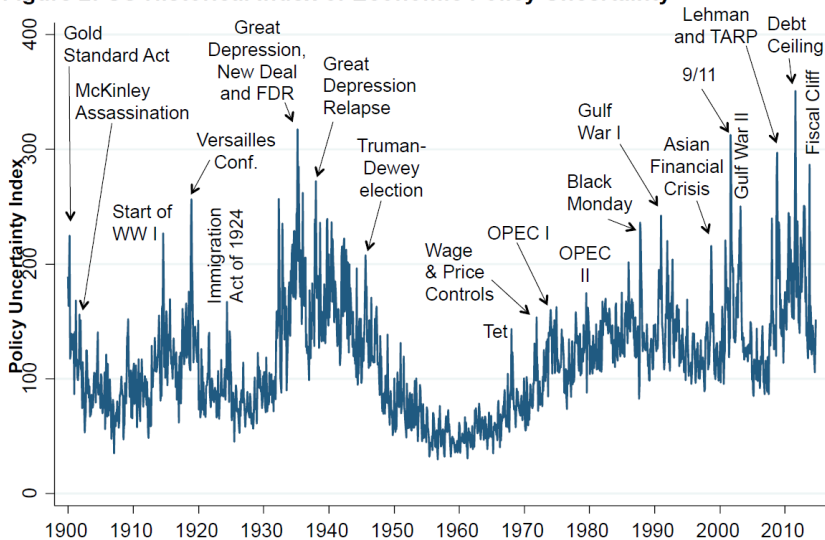
- Widely used in Political Science and Linguistic.
- Approach to text analysis: **deduction** and **abduction**
- Deduction:
 - Dictionary
 - Boolean
 - Easier to code, theory based
- Abduction:
 - Latent Dirichlet Allocation
 - Complex to code, data based

Boolean Technique

- Baker, Bloom, and Davis (2013), *Measuring Economic Policy Uncertainty*
 - They count the number of times that a word was repeated over the major US journals.
1. Article contains “uncertain” OR “uncertainty”, AND
 2. Article contains “economic” OR “economy”, AND
 3. Article contains “congress” OR “deficit” OR “federal reserve” OR “legislation” OR “regulation” OR “white house”.

Boolean Technique II

Figure 2: US Historical Index of Economic Policy Uncertainty



Dictionary method

- Start by defining a **list of key words** to capture content of interest.
- Then represent each document in terms of the (normalised) frequency of words in the dictionary.

$$\text{Dictionary} = \{\text{wage}, \text{labour}, \text{market}\} \quad (1)$$

$$\text{Share}_d = \frac{\#Wage + \#labour + \#Market}{\text{Totalwords}} \quad (2)$$

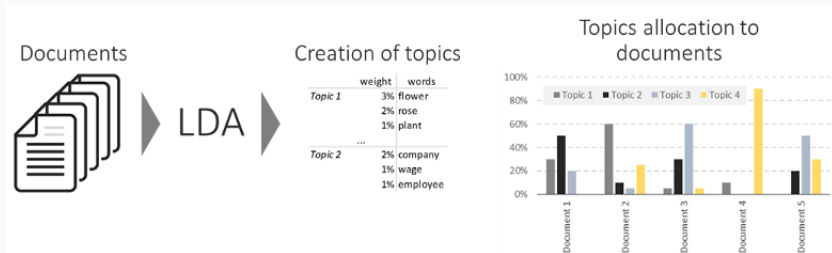
Dictionary method II

Tetlock (2007), *Giving Content to Investor Sentiment: The Role of Media in the Stock Market*.

- Used the Harvard IV-4, a particular dictionary that allows to classify in positive and negative the content of a text.
- He classifies *the Wall Street Journal* column “Abreast of the Market.”
- **Main finding:** pessimism predicts low short-term returns.
 - But...negative words in Harvard, not Negative in Accounting: tax, cost, capital, board, liability, foreign...
 - The main cons is to choose a bad dictionary for your analysis.

Latent Dirichlet Allocation

- LDA (short for Latent Dirichlet Allocation) is an unsupervised machine-learning model that takes documents as input and finds topics as output.



LDA: pros and cons

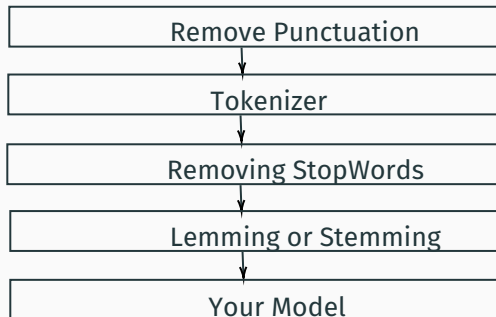
Pros:

- Works properly for long documents and large amount
- Usually fast to run

Cons:

- It needs human interpretation
- You cannot influence topics
- You should **normalize the text**.

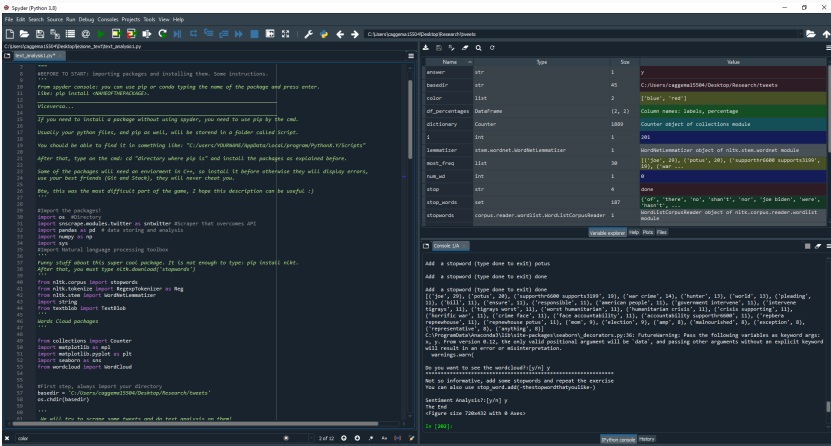
Text-Normalization Basic Algorithm



Python?

- Beginner-Friendly, most of the commands are just English words.
- Open source, thousands of developers and free information spread all over the world
- A really nice line on your CV.
- LBNL: can be useful for research and for working in the private sector.

Spyder: how does it look like?



Spyder: how does it look like? II

The image shows the Spyder Python IDE interface. A red arrow points to the 'Run' button in the toolbar. A red circle highlights the 'Variable Explorer' panel on the right, which displays a table of variables. Another red circle highlights the 'Console' panel at the bottom right, which shows the execution output. The main editor window displays a Python script for sentiment analysis.

Variable Explorer

Name	Type	Value
answer	str	1
baseurl	str	45
col	list	2
of_percentages	DataFrame	(2, 2)
dictinary	Counter	4
lemmatizer	WordNetLemmatizer	1
new_freq	list	30
num_md	int	1
stop	str	4
stop_words	set	1
stopwords	CorpusReader	1

Console

```
ADD a stopwords (type done to exit) potus
ADD a stopwords (type done to exit) done
ADD a stopwords (type done to exit) done
[('Joe', 20), ('Trump', 20), ('supporters600 supporters1109', 10), ('war crime', 14), ('hunter', 14), ('world', 11), ('pleading', 11), ('kill', 11), ('missile', 11), ('responsible', 11), ('american people', 11), ('government interference', 11), ('intervene', 11), ('highly secret', 11), ('worst humanitarian', 11), ('humanitarian crisis', 11), ('crisis accounting', 11), ('human rights war', 11), ('crime face', 11), ('face accountability', 11), ('accountability supporters600', 11), ('leaders', 11), ('house', 11), ('responsibility potus', 11), ('war', 9), ('election', 9), ('law', 9), ('unauthorised', 8), ('assault', 8), ('representation', 8), ('anything', 8)]
[('supporters600 supporters1109', 10), ('war crime', 14), ('hunter', 14), ('world', 11), ('pleading', 11), ('kill', 11), ('missile', 11), ('responsible', 11), ('american people', 11), ('government interference', 11), ('intervene', 11), ('highly secret', 11), ('worst humanitarian', 11), ('humanitarian crisis', 11), ('crisis accounting', 11), ('human rights war', 11), ('crime face', 11), ('face accountability', 11), ('accountability supporters600', 11), ('leaders', 11), ('house', 11), ('responsibility potus', 11), ('war', 9), ('election', 9), ('law', 9), ('unauthorised', 8), ('assault', 8), ('representation', 8), ('anything', 8)]
Do you want to use the package? (y/n) y
Yes
Help to download the new stopwords and report the exercise
You can also use stop_words.add(stopwords.get('potus'))
Sentiment Analysis: (p/n) y
Yes
File size 720412 with 4 Assoc
In (380):
```

Installing Packages

```
pip install <NAMEOFTHEPACKAGE>
```

Remove Punctuation

We are going to remove punctuation from our strings. Let's take Draghi's speech back!

Remove Punctuation II

And so we view this, and I do not think we are unbiased observers, we think the euro is irreversible. And it's not an empty word now, because I preceded saying exactly what actions have been made, are being made to make it irreversible. But there is another message I want to tell you. Within our mandate, the ECB is ready to do whatever it takes to preserve the euro. And believe me, it will be enough.



And so we view this and I do not think we are unbiased observers we think the euro is irreversible And it's not an empty word now because I preceded saying exactly what actions have been made are being made to make it irreversible But there is another message I want to tell you Within our mandate the ECB is ready to do whatever it takes to preserve the euro And believe me it will be enough

Remove punctuation: how to do it?

Removing Punctuation

```
1 import string
2 def remove_punctuation(text):
3     no_punct= "".join([c for c in text if c not in string.punctuation])
4     return no_punct
5
6 nopunct= remove_punctuation(text)
```

['and', 'so', 'we', 'view', 'this', 'and', 'i', 'do', 'not', 'think', 'we', 'are',
'unbiased', 'observers', 'we', 'think', 'the', 'euro', 'is', 'irreversible', 'and',
'it', 's', 'not', 'an', 'empty', 'word', 'now', 'because', 'i', 'preceded', 'saying',
'exactly', 'what', 'actions', 'have', 'been', 'made', 'are', 'being', 'made', 'to',
'make', 'it', 'irreversible', 'but', 'there', 'is', 'another', 'message', 'i', 'want',
'to', 'tell', 'you', 'within', 'our', 'mandate', 'the', 'ecb', 'is', 'ready', 'to', 'do',
'whatever', 'it', 'takes', 'to', 'preserve', 'the', 'euro', 'and', 'believe', 'me', 'it',
'will', 'be', 'enough']

Tokenizer application:

Tokenizer

```
1 from nltk.tokenize import RegexpTokenizer as Reg
2 tokenizer= Reg(r'\w+')
3 token= tokenizer.tokenize(text)
```

- We have some words that are not useful for our analysis. Like "The" or "So".

... little note, you have to download the stopwords running in the console:

```
nltk.download('stopwords')
```

Stopwords

```
1  from nltk.corpus import stopwords
2  stop_words = set(stopwords.words('english'))
3  def remove_stopwords(text):
4      words = [word for word in text if not word in stop_words]
5      return words
6  nostop= remove_stopwords(text)
7  print(nostop)
```

Stopwords II

- We have some words that are not useful for our analysis. Like "The" or "So".

['view', 'think', 'unbiased', 'observers', 'think', 'euro', 'irreversible', 'empty', 'word', 'preceded', 'saying', 'exactly', 'actions', 'made', 'made', 'make', 'irreversible', 'another', 'message', 'want', 'tell', 'within', 'mandate', 'ecb', 'ready', 'whatever', 'takes', 'preserve', 'euro', 'believe', 'enough']

- We reduced basically the text to a set of keywords.

Lemming or Stemming?

- **Stemming** and **Lemmatization** are **Text Normalization** (or Word Normalization) techniques in the field of *Natural Language Processing* that are used to prepare text, words, and documents for further processing
- The **goal** of both is to reduce inflectional forms and sometimes derivationally related forms of a word to a common base form.

Stemming:

- Stemming is the process of reducing inflection in words to their root forms such as mapping a group of words to the same stem.
- Stem (root) is the part of the word to which you add inflectional (changing/deriving) affixes such as (-ed,-ize, -s,-de,mis).
- For English: Porter-Stemmer or Lancaster-Stemmer

In brief...

- Playing, plays, played → play
- Am, are, is, were → be
- Car, cars, car's, cars' → car

Stemming: Porter-Stemmer application

```
1  #import packages
2  from nltk.tokenize import sent_tokenize, word_tokenize
3  from nltk.stem import PorterStemmer
4
5  porter = PorterStemmer()
6  def stemSentence(sentence):
7      token_words=word_tokenize(sentence)
8      token_words
9      stem_sentence=[]
10     for word in token_words:
11         stem_sentence.append(porter.stem(word))
12         stem_sentence.append(" ")
13     return "".join(stem_sentence)
14
15 stemmingdone= stemSentence(text)
16 print(stemmingdone)
```

Stemming: porter-stemmer result

and so we view thi , and i do not think we are unbias observ , we think the euro is irrevers . and it ' s not an empti word now , becaus i preced say exactli what action have been made , are be made to make it irrevers . but there is anoth messag i want to tell you . within our mandat , the ecb is readi to do whatev it take to preserv the euro . and believ me , it will be enough

- Do we need a more stronger stem? → **Lancaster!**
- **Be carreful**: over-stemming can lead to meaningless words.

Lancaster-stem: application

f

```
1  from nltk.stem import LancasterStemmer
2  lancaster=LancasterStemmer()
3
4  def stemLancaster(sentence):
5      token_words=word_tokenize(sentence)
6      token_words
7      stem_sentence=[]
8      for word in token_words:
9          stem_sentence.append(lancaster.stem(word))
10         stem_sentence.append(" ")
11     return "".join(stem_sentence)
12
13
14  stemmingstronger= stemLancaster(text)
```

and so we view thi , and i do not think we ar unbias observ , we think the euro is irrevers . and it ' s not an empty word now , becaus i prec say exact what act hav been mad , ar being mad to mak it irrevers . but ther is anoth mess i want to tel you . within our mand , the ecb is ready to do whatev it tak to preserv the euro . and believ me , it wil be enough

- ***Lemmatization***: refers to the use of a vocabulary and morphological analysis of words and to return the base or dictionary form of a word, which is known as the **lemma**.
- Lemmatization looks at surrounding text to determine a given word's part of speech, it **does not categorize** phrases.

- How to lemmatize?
 - It uses **Part-of-Speech (POS)** tagging to identify the grammatical class (part-of-speech) to which each word belongs –nouns, pronouns, verbs, etc. – and convert them into their base form.
 - ‘**Saw**’ tagged as a verb would become ‘**see**’ but not when tagged as a noun.

Lemming: application II

```
1 #import and download the needed packs.
2 from nltk.stem import WordNetLemmatizer
3 #Define the function:
4 def wordlemma(text):
5     lem_text = [lemmatizer.lemmatize(i) for i in text]
6     return lem_text
7
8 text= wordlemma(pre-lemmatext)
```

- Coming back to Draghi's Speech.

['view', 'think', 'unbiased', 'observer', 'think', 'euro', 'irreversible',
'empty', 'word', 'preceded', 'saying', 'exactly', 'action', 'made', 'made',
'make', 'irreversible', 'another', 'message', 'want', 'tell', 'within',
'mandate', 'ecb', 'ready', 'whatever', 'take', 'preserve', 'euro', 'believe',
'enough']

What do we do now?

- We try to create a dummy, that indicates if a word is in the string or not.

Extracting information

```
1 df = pd.DataFrame(columns=['text'])
2 df = df.append({'text': stemmingdone}, ignore_index=True)
3 df = df.append({'text': "text analysis is cool!"}, ignore_index=True)
4
5 df['dummy'] = np.where(
6     df['text'].str.contains('euro'), 1, 0)
```

Final Dataset:

df - DataFrame

Index	text	dummy
0	and so we view thi , and i do not think we are unbias obs...	1
1	text analysis is cool!	0

Second Part 😊

We go directly on Spyder

- Open *text_analysis1.py*
- What we will do?
 - Scraping, cleaning, visualizing and classifying.

What do we need?

import packages

```
1 import os #Directory
2 import snsrape.modules.twitter as sntwitter #Scraper that overcomes API
3 import pandas as pd # data storing and analysis
4 import numpy as np
5 import sys
6 #Import Natural language processing packages
7 from nltk.corpus import stopwords
8 from nltk.tokenize import RegexpTokenizer as Reg
9 from nltk.stem import WordNetLemmatizer
10 import string
11 #Sentiment Analysis
12 from textblob import TextBlob
13 #Words counter
14 from collections import Counter
15 #graphs
16 import matplotlib as mpl
17 import matplotlib.pyplot as plt
18 import seaborn as sns
19 #Words Cloud package
20 from wordcloud import WordCloud
```

Set your directory:

```
1 Set your directory, where you want to store your results.  
2 basedir = 'Your Directory'  
3 os.chdir(basedir)
```

What is web scraping?

Web-Scraping is a technique that allows us to extract data from the internet. The main package is: BeautifulSoup.

In this lesson we will talk about: : **snsrape**.

How snsrape works?

- You can use a keyword or a user's name: : like @POTUS to scrape all of the tweets that he has published.
- Other filters are written into the code and online materials.

What is web scraping? II

```
1   Creating list to append tweet data.
2   tweets_list = []
3   # filter: min_faves:10 OR min_retweets:10
4   # time filter: since:2012-01-01 until:2016-01-01
5   # Using TwitterSearchScraper to scrape data and append tweets to list
6
7   for i,tweet in enumerate(sntwitter.TwitterSearchScraper('biden min_retweets:1
    ↳ lang:en').get_items()):
8       # Use a break to limit your scrape
9       if i>200:
10          break
11          tweets_list.append([tweet.date, tweet.id, tweet.content, tweet.user.username,
    ↳ tweet.retweetCount, tweet.likeCount])
12          tweets_df = pd.DataFrame(tweets_list, columns=['Datetime', 'Tweet Id', 'Text',
    ↳ 'Username', 'retweet','likes'])
13          # Creating a dataframe from the tweets
14          tweets_df.to_csv('lecture_text_analysis.csv', sep = ';')
```

First scraped dataset

Index	Datetime	Tweet Id	Text	Username	retweet	likes
0	2022-04-04 10:06:14+00:00	1510921679828144134	Biden is compromised!!! Are there no Dem Co...	BradhamShannon	1	0
1	2022-04-04 10:05:19+00:00	1510921449384652803	Leht <pre>pre>6fe5n/0JXFR8g3...</pre>	DeanPorrazzo	1	1
2	2022-04-04 10:04:49+00:00	1510921321701646338	NEW: Secret Service paying over \$30K per mo...	Santucci	1	0
3	2022-04-04 10:04:18+00:00	1510921194408751107	@GOP Sure he was! He was there when he and ...	Hqdepot1	1	1
4	2022-04-04 10:02:20+00:00	1510920695647113218	More secret migrant flights fanning out fro...	softwarnet	2	1
5	2022-04-04 10:01:20+00:00	1510920443858984966	Biden is compromised!!! Are there no Dem Co...	BradhamShannon	2	1
6	2022-04-04 10:01:00+00:00	1510920362120339459	MITOSINGURE"20"1AULkbcAS"6/"HID"IN"20"04...	FitwiDesta	2	0
7	2022-04-04 10:00:54+00:00	1510920337894133763	Compromised Biden Cannot Lead America or th...	GarysBlues	3	8
8	2022-04-04 10:00:16+00:00	1510920175822028800	Biden is compromised!!! Are there no Dem Co...	BradhamShannon	1	1
9	2022-04-04 10:00:12+00:00	1510920160282038275	S.Korea president-elect's team visits U.S. ...	ReutersAsia	1	2
10	2022-04-04 10:00:10+00:00	1510920154271686658	Decreased supply played a bigger role in dr...	MacrobondF	1	2
11	2022-04-04 10:00:00+00:00	1510920110764027909	'THIS HAS TO STOP': @TikkiHaley says Biden'	FoxBusiness	1	4
12	2022-04-04 10:00:00+00:00	1510920109585543169	A team of foreign policy and security advis...	AlArabiya_Eng	1	1
13	2022-04-04 10:00:00+00:00	1510920109526851586	'THIS HAS TO STOP': @TikkiHaley says Biden'	FoxNews	7	52
14	2022-04-04 09:59:56+00:00	1510920095207690248	Biden is up to his neck in Ukraine.	x_x_0x_0_x	1	3
15	2022-04-04 09:59:50+00:00	1510920069676630020	@kirsincongress I can't see the west helpin...	alec_trusch	1	0
16	2022-04-04 09:59:33+00:00	1510919995613540352	President Joe Biden's son Hunter Biden is ...	Editorial_Write	2	1
17	2022-04-04 09:58:50+00:00	1510919849316392960	@GOPChairwoman @Foxnew @GOP rate from 3,11...	Hqdepot1	2	2
18	2022-04-04 09:58:43+00:00	1510919786485919744	#SupportHR6608 #SupportS3199	ShahiHabesha	1	0
19	2022-04-04 09:58:26+00:00	1510919714054189058	Is it just me or isn't it odd that Hunter B...	Marston4ca42	8	9
20	2022-04-04 09:58:10+00:00	1510919650116349956	@independentob10 @mikememoli @DaikonForTurn...	CamdenBlueRidge	1	1
21	2022-04-04 09:57:57+00:00	1510919595703644164	As @POTUS said "Time & again history s...	usosce	2	3

What are we going to do with this data?

- Apply text-normalization
- Study the frequencies
- Visualize a Word Cloud
- Easy sentiment analysis and visualization.
- → Python!

What is a Wordcloud?

- WordCloud is an useful tool to display e the most frequent words in your string.
- Using python you can allow it to display the most frequent bi-grams as well.
- Easy: the bigger is the word,the more frequent it is.

What is TextBob?

- Easy sentiment analysis algorithm. It gives a polarize sentiment (Neg Vs Pos, usually when we get zero we say it is neutral)
- Works pretty well with short text.
- Not specific for a particular topic.
- Built using movie reviews.

Thank you for your attention 😊

References:

- Tetlock, Paul C. "Giving content to investor sentiment: The role of media in the stock market." *The Journal of finance* 62.3 (2007): 1139-1168.
- Baker, Scott R., Nicholas Bloom, and Steven J. Davis. "Measuring economic policy uncertainty." *The quarterly journal of economics* 131.4 (2016): 1593-1636.
- Bholat, David, et al. "Text mining for central banks." Available at SSRN 2624811 (2015).
- Github folders and hints:
 - <https://github.com/openai/>
 - <https://github.com/JustAnotherArchivist/snsrape>
 - <https://github.com/slوريا/TextBlob>